

From Struggle to Enjoyment: Investigating ICH Digital Inheritance Through the Lens of Vocational Education

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Figure 1: We identify challenges in ICH vocational education and propose intelligence and digital solutions to transform learning experience from struggle to enjoyment through personalized, flexible, and engaging strategies.

Abstract

The preservation of Intangible Cultural Heritage (ICH) relies on the continued inheritance of craftsmanship, primarily facilitated by vocational education. However, most digital technologies for ICH emphasize public engagement and cultural promotion rather than comprehensive skill-building, making it unclear how best to support practitioner training. Through interviews with 10 embroidery practitioners, this study identifies three main vocational education challenges: high learning barriers, rigid teaching models, and societal constraints. The lengthy training cycle, repetitive practice, and frequent mistakes discourage learners, while traditional master-apprentice methods lack adaptability. Meanwhile, societal prejudice and economic uncertainty reduce the allure of ICH careers. To address these issues, we propose intelligent learning strategies that improve accessibility, personalize skill development, and offer better career prospects. Our findings provide empirical evidence for

the HCI community and underscore the necessity of more inclusive, sustainable vocational education frameworks for ICH.

CCS Concepts

• **Human-centered computing** → *Interactive systems and tools; Empirical studies in HCI.*

Keywords

ICH digital inheritance, vocational education, qualitative study

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1 Introduction

Protecting and inheriting Intangible Cultural Heritage (ICH) is essential for preserving human civilization's diversity [23]. ICH transmission depends on generational relay, and without inheritors or practitioners, these skills and knowledge risk fading away [28]. Thus, vocational education aimed at training professionals is key to safeguarding ICH.

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In HCI, many studies leverage digital technologies for ICH education, focusing on popular science and engaging experiences. Examples include using AR, VR, serious games, and open-source hardware to convey ICH's historical and cultural knowledge while enhancing the enjoyment of hands-on creation [20, 26, 30, 34]. However, such approaches fall short in vocational education, as they often neglect the systematic training and prolonged practice needed to master craftsmanship. Practitioners require extensive hands-on experience to hone advanced skills and secure market-based income for sustainability. Moreover, although studies note young people's disinterest in ICH careers[37], few investigate the root causes or explore how emerging technologies can tackle these challenges to advance ICH vocational education. Therefore, this study poses two research questions:

- What factors hinder individuals from becoming professional ICH practitioners?
- How can intelligent technologies help address these challenges to advance ICH vocational education?

We conducted interviews with 10 teachers and students specializing in ICH embroidery to explore the challenges in vocational education. Embroidery, with its solid inheritance foundation and stable social demand, still faces low public interest in learning, which hinders its transmission and development. This issue reflects a broader challenge in ICH preservation.

Our research found that the three key challenges in ICH inheritance are high learning barriers, rigid vocational education models, and societal biases that hinder both learner engagement and economic sustainability. First, learning embroidery has high entry barriers, such as long training periods and repetitive processes, requiring both systematic training and long-term commitment. Second, traditional vocational education often lacks personalized, student-centered approaches and fails to meet diverse needs. Lastly, societal biases against ICH professions, coupled with low economic returns, further reduce learners' enthusiasm. To address these, we propose design strategies that integrate intelligent technologies to offer low-threshold, personalized, engaging, and flexible learning experiences. This would allow more people to engage in ICH learning anytime, anywhere, expanding its inheritance. For professional monetization, we advocate for comprehensive societal support, including job matching based on skill levels and incorporating commercial knowledge to help learners turn their skills into economic value.

The contributions of this late-breaking work are as follows: First, it is the first to approach ICH inheritance from a vocational education perspective, highlighting the pain points and challenges in ICH vocational education. Second, it proposes design strategies that integrate intelligent technologies and outlines a future research roadmap and technical framework, offering insights for the advancement of ICH vocational education and related studies.

2 Background and Related work

In the field of HCI, numerous studies on ICH education employ technologies such as multimedia[27][33], software applications[2], serious games[20, 26, 30, 34], and VR/AR[34, 36, 39] to disseminate the historical and cultural knowledge of intangible cultural heritage

to the public. For instance, Tan et al. developed the CAI WebAR system to demonstrate the production process of Cantonese porcelain in classrooms, significantly enhancing interactivity and engagement in learning[36]. Endow et al. analyzed different forms of tangible craft tutorials and advocated for a shift from traditional tutorial formats to more interactive and context-sensitive tutorials[9].

In craft workshops and innovative fabrication contexts, researchers have utilized open-source hardware, such as Arduino, servos, and LED strips, to guide learners in creating dynamic or interactive ICH artifacts[14, 25, 28]. Examples include dynamic Hair Monkey crafts[23], interactive Chilean arpillera paintings[11], embroidery with display functions[8], and ceramics with gesture recognition capabilities[40]. Additionally, assistive crafting systems have been developed to help beginners create a variety of artifacts with ease, such as honeycomb smocked handbags[7], slab-based ceramic pieces[13], woven tapestries[1, 10, 12, 19], and tangible embroideries[3, 15, 17, 31]. While these approaches enhance learner engagement, they are often limited to one-off, short-term experiences and fail to address the structured learning needs emphasized in vocational education.

A few studies focus on using digital technologies to improve knowledge transfer and teaching efficiency in classroom settings[4]. For example, Ye et al. used eye-movement modeling examples (EMME) to visualize the visual patterns of craft experts, aiding learners in understanding tacit procedural knowledge[38]. Similarly, intelligent tutoring systems like InStitches provide targeted exercises to help users acquire specific skills[21]. Jones et al. discussed the educational challenges in embroidery using examples of textile mending[16] and goldwork embroidery[18]. However, the students' experiences have not been fully explored in these works. Although these technologies show promise in facilitating knowledge transfer, they remain limited in supporting the long-term mastery of complex craft skills and professional development.

3 Methodology

3.1 Participants

We recruited 10 participants (8 female, 2 male, ages 21–50) with diverse expertise in Su embroidery, Beijing embroidery, and other regional forms. They included inheritors, workshop apprentices, university students and other practitioners in embroidery industry. Table 1 outlines their demographics, spanning those with under one years of experience to over a decade in heritage practice.

3.2 Procedures

We conducted face-to-face interviews in Suzhou and Hangzhou, China when schedules and locations permitted. In order to further understand the real needs, the research team went to the master studios in Suzhou and Hangzhou to conduct offline research. For participants in other cities, we conducted online interviews. We explore the needs and challenges of computer-assisted embroidery learning from both learners' and instructors' perspectives. For learners, we focus on skill acquisition, competency assessment, learning processes, challenges and strategies, as well as their suggestions for intelligent tutoring tools. For instructors, we examine vocational training, the cultivation of successors, teaching methods, difficulties, and the use of technological tools. Instructors also share their own

learning experiences or reflect from a student's perspective. After gaining an initial understanding, we propose ideas for computer-assisted embroidery learning tools and explore their possibilities together. When interesting insights emerge, we follow up with questions about more details and specific examples. Each interview lasted about an hour, and participants received compensation of approximately \$7.

3.3 Data analysis

We employed thematic analysis [5, 24] to process the data. All interviews were conducted in Mandarin, recorded via conferencing software, and transcribed into text. Each researcher independently read and coded the interview transcripts. During this process, we stored interview data in cloud documents and used mind-mapping tools for qualitative analysis. We frequently revisited the conversations and details to deepen our understanding of the data. Through open coding, each researcher generated an initial set of codes, which were then discussed among the authors in meetings. During this process, themes such as impatience, having high expectations but low ability, inconsistent evaluation standards, and economic returns emerged. Then, we organized these themes into categories and subcategories. Ultimately, we identified three key themes: the high barrier to entry in embroidery learning poses learning challenges, existing teaching models fail to accommodate diverse learning needs, and the social ecosystem of embroidery constrains industry development.

4 Findings

We will elaborate the challenges in embroidery vocational education through three perspectives: the nature of embroidery skills, the vocational education model, and societal support.

4.1 The High Learning Threshold of Embroidery Brings Learning Challenges

Many participants mentioned significant challenges in learning embroidery, often leading to discouragement and a desire to quit. As P8 said: *"I want to give up every day. What can you do?"* Upon further analysis, we identified the following challenges.

4.1.1 The Monotony and Long Learning Curve of Embroidery Discourages Learners. Learning embroidery involves a wide variety of stitches and an extensive body of knowledge, making it challenging to fully grasp within a short period. As P2 mentioned: *"Even if you master several stitches in class, it's easy to forget them later."* Therefore, many learners view learning embroidery techniques as a monotonous process that requires long-term dedication. For example, P8 said: *"The time to get started is too long. Even after studying for several years, you may not produce a good piece."* P1 also mentioned that: *"Compared to painting, mastering embroidery takes much longer."* This low-reward, high-investment learning model leads many learners to feel frustrated and want to quit early on.

4.1.2 Frequent Errors and Costly Rework in Learning Lead to Frustration. In embroidery learning, frequent mistakes force learners to undo and redo stitches, a time-consuming process requiring precise control to avoid damage. This repetitive cycle leads to significant frustration. As P3 said: *"Stitching and undoing, over*

and over again—it's so frustrating when undoing stitches that you just feel like giving up!"

Learners often need to rip out stitches, primarily due to difficulties with color matching (P3, P4, P8, P10). As P8 explained, they frequently unpick and adjust their work to align with the original design: *"I sit down with one color, and when I stand up, it looks like a completely different color. So, I rip it out and re-stitch."* The complexity of the subject matter also adds to the need for revisions. P10 told us: *"I really want to learn to stitch animals, but they're much harder. They're on a whole different level compared to flowers and birds."* Additionally, learners often modify their work to resolve composition issues, as P9 noted: *"The relationship between light and shadow, and the contrast between fullness and emptiness, is similar to painting. For those who haven't studied art, this can be quite challenging."*

4.1.3 The Contrast Between Embroidery's Simplicity in Appearance and Its Difficulty in Practice Demotivates Learners.

Although finished embroidery pieces may look easy to create, many participants shared that the actual stitching process is far more challenging, requiring significant skill and patience. P4 explained: *"Everyone sees the double-sided embroidered fan and thinks 'It's beautiful, I want to stitch this'. I tell them they can only handle simplified patterns, but they reply, 'That looks too simple, not what I imagined.'"* P10 added: *"I've been learning to embroider animals for eight years and still can't do it like the teacher. I can visualize it perfectly, but my hands just won't cooperate."* This paradox of "hands and mind not working together" frustrates many learners, causing them to lose interest in the creative process.

4.2 The Teaching Model Cannot Accommodate Diverse Learning Needs

Learner dropout rates are closely linked to both embroidery instructors and the learners themselves. P4 shared: *"Recruitment is quite difficult. Fewer and fewer people are interested in these slow crafts."* Upon further analysis, we found that the teaching model's failure to accommodate diverse learning needs has led many learners to struggle in achieving their goals within the current embroidery system.

4.2.1 Teaching Perspective: Emphasizing Long-Term Apprenticeship Training. Instructors often focus on "serious teaching" of embroidery, offering long-term or even lifelong instruction. Currently, embroidery education lacks a systematic and standardized training program. Teaching content is often driven by the production requirements of commercial orders. As P5 mentioned: *"If a client orders a batch of magnolia flowers, we focus on practicing magnolia flowers intensively. For the same design, I make you stitch it 10, 15, or 20 times before it's considered acceptable."*

In this unstandardised, long-term training model, instructors emphasize that learning embroidery requires extensive practice (P3, P4), and a deep passion and focus on the craft (P10). P4 said: *"If you want to become a professional embroiderer, you must be ready to dedicate yourself and prepare for a long, enduring battle."* P10 shared: *"I believe that to truly enjoy embroidery, you must deeply love it, not just like it."*

4.2.2 Student Perspective: The Diverse Learning Needs. In today's fast-paced society, few learners are willing to dedicate their

Table 1: Demographic information of 10 participants.

No.	Gender	Age	Embroidery Styles	Occupation	Years of Experience
P1	M	36	Suzhou embroidery	Renowned inheritor	More than ten years
P2	F	21	Bead embroidery	University student	Less than one year
p3	F	36	Suzhou embroidery	Practitioner	More than ten years
P4	F	35	Suzhou embroidery & Gansu's Long embroidery	Inheritor	More than ten years
P5	M	38	Beijing embroidery	Inheritor	More than ten years
P6	F	20	Zhejiang embroidery & Bead embroidery	Univeristy student	Less than one year
p7	F	30	Suzhou embroidery	Embroidery workshop apprentice	Apprentice for three years
p8	F	26	Suzhou embroidery	Embroidery workshop apprentice	Apprentice for five years
p9	F	24	Suzhou embroidery	Embroidery workshop apprentice	Apprentice for three years
P10	F	50	Hangzhou embroidery	Renowned inheritor	More than ten years

lives to embroidery. Embroidery vocational learners typically fall into two groups: arts and crafts students in higher education and workshop trainees, each with diverse learning needs. For instance, some focus on economic benefits. As P8 explained, *"Many people pursue this just to make a living, it's not as noble as it sounds."* Others have clear short-term goals, aiming to master basic embroidery techniques to incorporate into their other works. P1 noted, *"Some students are in fashion design programs and want to use embroidery in their creations. I've noticed a significant demand for this among Chinese students studying abroad."* There are also those who are drawn to the creative process of embroidery, seeing it as a source of emotional fulfillment. As P8 and P9 shared, *"I feel a great sense of accomplishment after completing embroidery. Creating my own work is also a way to motivate myself."*

4.2.3 The Single Teaching Model Excludes Diverse Learners.

Instructors follow the "apprenticeship" model when recruiting students, offering a single, traditional professional pathway. However, this one-size-fits-all approach doesn't meet the diverse needs of modern learners. P4 pointed out that many embroidery learners don't fit the current teaching model: *"Contemporary university students don't have the patience to go from start to finish."* For learners with different goals, instructors often attribute unadaptable outcomes to the learners themselves. P8 noted: *"Some people can't stick with it after a week or two of learning, and such people are mainly impatient."*

The single teaching model retains only those learners who can adapt to traditional methods, who are likely to become the educators preserving this approach in the future. However, this rigid inheritance model not only excludes learners who cannot adapt but also hinders the economic growth of the industry. This economic challenge further exacerbates the income difficulties faced by both instructors and learners. P5 shared: *"If you come from a poor family, we wouldn't even accept you here."* P8 added: *"If you come to learn embroidery and I tell you that I won't pay you for the first five years, how will you come?"*

4.3 Social and Industry Constraints in Embroidery Learning

We found that the development of the embroidery industry is constrained by multiple factors from the social environment. As a traditional craft, embroidery faces negative perceptions within the modern value judgment system.

4.3.1 Challenges in Aligning Traditional Embroidery with Modern Employment and Market Needs. After completing their training, people naturally hope to achieve good economic returns. However, the lack of innovation awareness and capability prevents them from competing with other low-cost, mass-produced products. As P5 said, *"Double-sided embroidered fans sold in live-streaming sessions cost only 30 to 40 yuan and are all North Korean embroidery, while our domestic hand-embroidered ones take two or three days to make."* Embroidery practitioners also face challenges in employment, often limited by traditional mindsets and lacking product design and commercialization knowledge (P4, P5, P6). As P6 pointed out: *"Many skilled embroiderers struggle to enter prestigious factories due to constraints from traditional mindsets."* P4 added: *"Many ICH inheritors aim to create and sell products but lack support from product designers."*

4.3.2 The Embroidery Industry's Exclusivity Limits Recognition of Professional Identity. The exclusivity of the embroidery industry is evident both externally and internally. Externally, it is primarily reflected in the high entry barriers, with many participants emphasizing the difficulties of breaking into the field. As P9 said: *"Embroidery takes three years just to get started."* However, P1 offered another perspective: *"Some say it takes decades to master embroidery, but this can sometimes act as a deterrent within the industry. When seniority becomes the standard for evaluation, many people lose interest in joining."* From an internal perspective, the exclusivity of the embroidery industry is also reflected in the narrow definition of professional identity for practitioners. P1 explained: *"The evaluation system excludes 80–90% of practitioners by requiring costly exhibition fees to advance, setting a high barrier for participation."*

4.3.3 Societal Bias and Negative Perceptions of the Embroidery Industry. Society generally holds a negative view of the

embroidery industry, seeing it as an "outdated" skill. This perception is even reflected among practitioners within the industry. P8 shared: *"Many families who do embroidery don't want their children to suffer the same fate."* This negative perception diminishes the motivation of practitioners to continue their work. P8 believes: *"Su embroidery is not in high demand in people's lives anymore, and over time, it will gradually be phased out."* Our research also uncovered a significant gender bias within the embroidery profession. During interviews, participants such as P7, P8, P1, and P5 often referred to practitioners as "female embroiders" in Chinese, without using inclusive terms like "embroiders".

5 Discussion and Design implications

In our previous study, using embroidery as a case study, we identified key challenges facing ICH vocational education, including high learning barriers, rigid teaching models, and societal constraints. Mastering these crafts requires long-term training, yet the traditional apprenticeship model struggles to accommodate the diverse needs of modern learners. Moreover, market and societal biases against ICH-related professions further exacerbate the difficulties of cultural transmission, hindering the long-term sustainability of embroidery as a viable career. Addressing these issues calls for innovative design strategies that leverage digital technology to lower learning barriers, enhance the learning experience, and improve industry adaptability.

When exploring the impact of digital technology on craft skill acquisition, prior research has highlighted the crucial role of tutorial formats and content presentation. For instance, Endow & Torres[9] found that different tutorial media—books, videos, and online resources—affect how learners acquire tacit knowledge. While video tutorials can reduce cognitive load for beginners through intuitive demonstrations, books and online guides provide more structured knowledge accumulation. However, these media often lack support for long-term skill development, overlooking the systematic training and deep expertise required in vocational education. Meanwhile, Moradi et al.[29] discovered that ceramic learners develop "material epochs" through prolonged interaction with materials, refining their understanding via repeated experimentation, adjustment, and creation. This insight suggests that ICH vocational education should not only lower entry barriers but also incorporate mechanisms that promote sustained skill accumulation, enabling learners to iteratively refine their craftsmanship rather than relying solely on one-off or fragmented tutorials.

In the following sections, we will build upon these insights to propose an intelligent, personalized, and sustainable design strategy for ICH vocational education, aiming to shift the learning experience from struggle to enjoyment, and to facilitate the smooth transition of more learners into the preservation and innovation of ICH craftsmanship.

5.1 Design Strategy: From Struggle to Enjoyment Learning experience

To bridge the gap between traditional ICH education and modern learning needs, we propose design strategies that integrate digital and intelligent technologies, offering a more accessible, adaptable, and career-oriented learning experience.

(1) Lower Barriers to Entry and Enhance Learning Enjoyment. Traditional ICH education often emphasizes complex techniques and high-level artworks, assuming "three years to start, ten years to master." This lengthy and challenging learning model discourages many learners. To address this, future research should focus on designing a curriculum with fine-grained stages, allowing learners to engage in tasks and receive positive feedback at each step, rather than waiting for a long training period to "graduate." This approach helps learners progressively build their skills while staying motivated and gaining practical experience at every stage. Additionally, interactive challenges, gamification, and smart tools (such as color-matching tools in embroidery) can enhance both learning enjoyment and efficiency.

(2) Catering to Learners' Diverse Needs with Personalized Adaptive Learning. Traditional ICH teaching models fail to meet the diverse needs of modern learners, often relying on fixed content and methods that do not accommodate individual differences. Therefore, we advocate for a personalized teaching approach. By analyzing learners' interests, characteristics, and skill levels, digital technologies can recommend the most suitable courses and learning paths, ensuring each learner receives tailored support. Meanwhile, teachers' knowledge and expertise can be made "opensource" to digital agents, allowing for better alignment between teaching styles and learner needs, thereby improving teaching efficiency and the overall learning experience.

(3) Adapting to Modern Needs and Enhancing the Career Prospects of ICH Learners.

For learners, the ultimate goal of vocational education is to be able to translate skills into actual career opportunities and reap financial rewards. Therefore, a nice "intelligent mentor" should help students master the ICH skills while promoting the expansion of their career development ideas. Intelligent mentor can help students integrate traditional embroidery with modern design applications and enhance their innovation ability through personalized recommendations and combining their interests and skills. At the same time, the mentor should match the corresponding practice opportunities according to the level of students and market demand, such as helping students receive projects in line with their ability. In this way, digital technology not only supports technical learning, but also helps students understand and adapt to market needs and gradually realize the process from skill acquisition to career monetization.

5.2 A Future Research Framework for Advancing ICH Vocational Education

In the era of artificial intelligence, we have the opportunity to leverage AI mentors to provide accessible, personalized, and flexible remote learning experiences, unrestricted by time and location. AI mentors are not intended to replace traditional master-apprentice relationships but rather to complement and extend them. By accommodating diverse learning needs, they can reach a broader audience and further drive the transmission of ICH craftsmanship. Based on computational education models[6, 22, 32, 35], we propose a framework (Figure 2) comprising a user interface, student model, teaching model, and domain model to build an Intelligent Tutoring System as an "AI mentor".

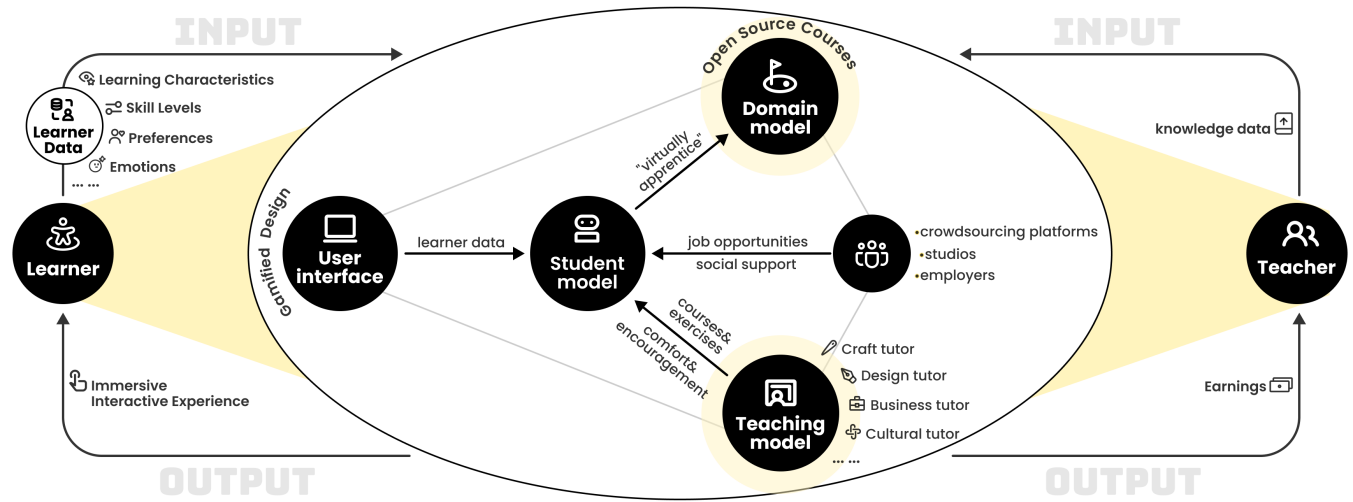


Figure 2: A future research framework for advancing ICH vocational education.

The student model uses a data-driven approach to capture learners' learning characteristics, tracking preferences, skill levels, and emotions, acting as a "digital avatar" of the learner.

The teaching model serves as a "learning mentor," using data from the student model to invoke intelligent agents that provide personalized and progressive learning guidance. For example, it recommends courses and exercises based on the learner's level and offers comfort and encouragement when the learner feels discouraged.

The domain model creates a digital avatar by open-sourcing the mentor's course and knowledge data. Learners can "virtually apprentice" with mentors, and mentors will receive a share of earnings and rewards for each student they accept. This model not only increases mentors' income but also provides more mentor options for learners.

The user interface attracts learners with a gamified design, offering an immersive, interactive experience to help develop learning habits. The system should also connect to crowdsourcing platforms, studios, and employers, intelligently matching learners' progress and abilities with job opportunities and providing social support.

5.3 Conclusion

In Intangible Cultural Heritage (ICH) education, many HCI studies focus on popular science and experiential learning, often overlooking the vocational education required to master craftsmanship through systematic training and prolonged practice. However, professional practitioners play a crucial role in the transmission of ICH skills. This study, using embroidery as an example, examines the high learning barriers, limitations of teaching models, and societal constraints in ICH vocational education. It proposes design strategies leveraging digital and intelligent technologies to lower entry barriers, provide personalized learning experiences, and enhance social support. Additionally, it outlines a future research framework to further advance ICH vocational education. By approaching ICH education from a vocational perspective, this work seeks to provide

new insights for both ICH inheritance research and the HCI community, contributing to the broader discourse on technology-enhanced craftsmanship learning.

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